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The Editors
Artificial Intelligence in Medicine

Dear Editors,

We submit the research article "Separating decision-rule failures from representation failures in external validation of diabetic retinopathy grading models" for consideration in *Artificial Intelligence in Medicine*.

When a diabetic retinopathy grading model fails on a new population, the failure is usually attributed to the learned representation and answered by changing the model. Ordinal grading models, however, convert scores into grades with a decision rule that is rarely examined. Across ten development models, four external populations, the reverse direction of transfer, two models fine-tuned on 28,096 EyePACS images and five training seeds, we show that the ranking of images largely transferred while the mapping from score to grade did not: the same models over-graded one population and under-graded another, so that a referral threshold fitted for 85% sensitivity reached 99.33% on one and 63.38% on the other. The same threshold fitted on EyePACS over-referred everywhere, which shows that the direction of the shift belongs to the population and the sign of a transferred threshold's error to the development labels. Refitting cut-points recovered part of the loss, but only where ranking survived and only with local labels; label-free adaptation did not help.

We believe the work suits the journal for three reasons. It contributes evaluation methodology for medical AI: a decomposition of external performance into per-threshold discrimination and the score-to-grade mapping, and a matched decision rule for comparing models, with its record of changed and unchanged conclusions, including a published model whose reported advantage did not survive it. It gives a deployment rule that holds for any model: the risk that a locally set referral threshold misses its sensitivity floor depends on the number of referable cases labelled, follows a distribution-free form, and translates into images at any screening prevalence. And it is transparent about its own evidence: the analysis plans were committed to a public repository before the corresponding analyses, every outcome is reported with its verdict, falsified ones included, post-hoc analyses are labelled, and every number in the manuscript is generated from archived per-image predictions that are public with the code.

The manuscript has not been published and is not under consideration elsewhere. Parts of the development work form the basis of a Master's thesis at the University of Tehran. All data are public and de-identified. The authors declare no competing interests. The use of an AI assistant for code, analysis runs and text is described in the Methods and in the generative-AI declaration.

Sincerely,

Ahmadreza Nourozi
(on behalf of both authors)